

Chapter 6

Proportions and Similarity

Unit 2 — Triangles

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Key Vocabulary

Term	Definition
Ratio	Comparison of two quantities: $\frac{a}{b}$
Proportion	Equation stating two ratios are equal: $\frac{a}{b} = \frac{c}{d}$
Extremes	Outer terms of a proportion (a and d)
Means	Inner terms of a proportion (b and c)
Similar polygons	Same shape, proportional sides, congruent angles
Scale factor	Ratio of corresponding sides of similar figures
Dilation	Enlargement or reduction preserving shape

Cross-Products Property

In the proportion $\frac{a}{b} = \frac{c}{d}$, the cross products are equal:

$$ad = bc$$

Example: $\frac{x}{6} = \frac{5}{3}$

$$3x = 30 \Rightarrow x = 10$$

Properties of Proportions

Property	Statement
Reciprocal	If $\frac{a}{b} = \frac{c}{d}$, then $\frac{b}{a} = \frac{d}{c}$
Exchange Means	$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a}{c} = \frac{b}{d}$
Addition	$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a+b}{b} = \frac{c+d}{d}$

Definition: Similar Polygons

Two polygons are **similar** if:

1. All corresponding angles are **congruent**
2. All corresponding sides are **proportional**

$$\triangle ABC \sim \triangle DEF$$

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} = k \quad (\text{scale factor})$$

Postulate 6-1: AA (Angle-Angle) Similarity

If two angles of one triangle are congruent to two angles of another, the triangles are **similar**.

$$\angle A \cong \angle D \text{ and } \angle B \cong \angle E \Rightarrow \triangle ABC \sim \triangle DEF$$

Only 2 angles needed — the 3rd follows from the angle sum.

Theorem 6-1: SSS (Side-Side-Side) Similarity

If all three pairs of corresponding sides are **proportional**, the triangles are **similar**.

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} \Rightarrow \triangle ABC \sim \triangle DEF$$

Theorem 6-2: SAS (Side-Angle-Side) Similarity

If two pairs of corresponding sides are **proportional** and the **included angles are congruent**, the triangles are **similar**.

$$\frac{AB}{DE} = \frac{BC}{EF} \text{ and } \angle B \cong \angle E \Rightarrow \triangle ABC \sim \triangle DEF$$

Theorem 6-3: Triangle Proportionality Theorem

If a line is **parallel to one side** of a triangle and intersects the other two sides, it divides them **proportionally**.

$$DE \parallel BC \Rightarrow \frac{AD}{DB} = \frac{AE}{EC}$$

Theorem 6-4: Converse of Triangle Proportionality

If a line divides two sides of a triangle **proportionally**, it is **parallel** to the third side.

$$\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow DE \parallel BC$$

Theorem 6-5: Triangle Midsegment Theorem

The segment connecting the **midpoints** of two sides of a triangle:

1. Is **parallel** to the third side
2. Is **half the length** of the third side

$$DE \text{ is midsegment} \Rightarrow DE \parallel BC \text{ and } DE = \frac{1}{2}BC$$

Example — Midsegment

In $\triangle ABC$, D is the midpoint of $\overline{AB}$ and E is the midpoint of $\overline{AC}$.

If $BC = 14$, find DE .

$$DE = \frac{1}{2}(14) = 7$$

Theorem 6-6: Proportional Parts (Parallel Lines)

If three or more parallel lines intersect two transversals, they divide the transversals **proportionally**.

$$\ell_1 \parallel \ell_2 \parallel \ell_3 \Rightarrow \frac{AB}{BC} = \frac{DE}{EF}$$

Theorem 6-7: Angle Bisector Proportionality

An **angle bisector** of a triangle divides the opposite side proportionally to the adjacent sides.

$$\overrightarrow{AD} \text{ bisects } \angle A \Rightarrow \frac{BD}{DC} = \frac{AB}{AC}$$

Similar Triangles — Perimeters and Areas

Property	Scale Factor k
Perimeters	Ratio = k
Areas	Ratio = k^2

Example: Scale factor = $\frac{3}{2}$

- Perimeter ratio = $\frac{3}{2}$
- Area ratio = $\frac{9}{4}$

Chapter 6 — Theorem Summary

#	Name	Key Idea
Post. 6-1	AA Similarity	2 $\angle$ s congruent $\rightarrow$ similar
Thm 6-1	SSS Similarity	All sides proportional $\rightarrow$ similar
Thm 6-2	SAS Similarity	2 sides prop. + included $\angle \cong$
Thm 6-3/4	Triangle Proportionality	Parallel line $\leftrightarrow$ proportional sides
Thm 6-5	Midsegment	$\frac{1}{2}$ length, parallel to base
Thm 6-6	Parallel Lines	Divide transversals proportionally
Thm 6-7	Angle Bisector	Divides opp. side proportionally